

**Gemini
Near
Infra-Red
Imager
and
Wavefront
Sensor**



Summary of the GNIRS OIWFS Mechanism Testing

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This document describes the tests which were performed to insure that the GNIRS OIWFS mechanisms were working correctly.

1 Introduction

The GNIRS OIWFS mechanisms were built using the same designs as the NIRI OIWFS mechanisms. The NIRI mechanisms had received extensive testing, both in the lab and on the telescope, so it was felt that the design was sound. For this reason, it was deemed unnecessary to perform extensive and time-consuming mechanism tests. It appeared sufficient to perform a basic test of all mechanism functions, with particular attention being paid to those elements which had proven to be problematic with the NIRI OIWFS.

1.1 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.2 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.3 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Filter Wheel

The filter wheel built from a design that has proven to be very reliable. For this reason, it was felt that it was acceptable to test the mechanism at room temperature, without cooling it to its operating temperature.

Testing was done using the actual interface boards and stepper-motor driver modules that will be used in GNIRS. Custom interface cables were used in place of the cables which will be used in the dewar; the other cables, which will be external to the dewar, were the cables which are to be used with GNIRS. The mechanism was configured as it was expected to be used, including the Hall-effect sensors.

The standard OIWFS datum / diagnose function was then used, with the NIRI calibration constants. It was expected that the datum function would work correctly, but that the diagnose function would fail until the system had been calibrated. It was found that the direction of rotation of the motor was the reverse of NIRI. It is believed that the NIRI motors were inconsistently wired. Since all of the GNIRS motors were wired identically, it was felt that it was preferable not to reverse the direction of motor motion to match NIRI. Instead, all calibration constants were reversed.

With the corrected calibration constants, the filter wheel was able to datum properly, but failed the diagnosis phase. This was expected, as the normal calibration constants are sensitive, and need to be regenerated for each mechanism. The low-level calibration constants are well-defined by the mechanism. They were not changed, and were not expected to need to be changed.

The standard mechanism calibration procedure was then carried out, and the new constants were verified to be reasonable, and then were loaded into the database. The datum / diagnose function was again performed, and was completed with no problems. The datum / diagnose function was then executed a few times, with no failures detected.

The mechanism was then moved to each filter position in sequence. The mechanism reached each position and the Hall-effect sensor reading confirmed that the mechanism was in the correct position.

Finally, the OIWFS cycle test was started. The mechanism was run through 30 test cycles. No errors were detected during the cycle test.

3 Focus Stage

The filter wheel built from a design that has proven to be reliable. For this reason, it was felt that extensive testing was unnecessary, but that cryogenic operation needed to be verified.

Testing began with the mechanism sitting in an upright position. The mechanism was moved to the center of travel. Some slight misalignment of the mechanism were discovered and corrected at this time. The home position was found to be slightly closer to the center of travel of the mechanism than NIRI. This has the possibility of introducing slightly more inconsistency to the home position, because the mechanism is not pre-loaded at the home position. It

was felt that the home position was sufficiently distant from the central position to insure a repeatable home position.

The mechanism was centered, and moved back and forth from limit to limit several times. It was found that the mechanism moved freely, with no apparent problems. No evidence of motor stalling — a common problem with this type of mechanism, particularly near the limits of travel — was detected.

The low-level calibration constants were then adjusted to match this mechanism. This was expected to be necessary, since the constants are dependent on very fine positioning of the Hall-effect sensors. The limits of travel were adjusted, also expected, as the limits depend on the orientation of the main screw. The mechanism was datumed, and performed as expected. The normal calibration was then carried out. The datum / diagnose test was then performed several times, with no failure. Finally, a cycle test was carried out for about 30 cycles. The mechanism passed with no problems.

The mechanism was then cooled to its operating temperature. The mechanism was unable to datum itself. Further inspection showed that the home position had moved slightly as the system had cooled. This was not a problem with Niri, though it was a problem with a similar Niri mechanism. At that time, it was decided to resolve the problem by changing some of the low-level calibration constants, depending on the temperature. This was found to be necessary with the Niri science-channel focus stage, which has a similar design. There is no explanation of why this is necessary with the GNIRS OIWFS focus stage, but not with the Niri OIWFS focus stage. With this change to the software, the mechanism passed the datum / diagnose test several times. Finally, a cycle test was performed for about 30 cycles. No problems were discovered during the cycle test.

Finally, to verify that nothing had changed with the mechanism was warmed again. It was able to pass the datum / diagnose test and a cycle test of approximately 30 cycles.

At this time, it is possible to say that the GNIRS OIWFS mechanism passes its tests as well as the corresponding Niri mechanism did. However, optical testing is still required to insure that it is behaving as well as the Niri mechanism.

4 Gimbal Mirror

The gimbal mirror was built from a working design, so it was felt that it was unnecessary to perform the detailed testing that was required for the Niri mechanism. This mechanism is designed to very high precision, so optical testing was required.

The mirror was first calibrated while it was warm. The mechanism was found to move smoothly. The mechanism was datumed a few times, and found to return to the same home position (as determined using the Hall-effect sensors). The mechanism was then calibrated and passed a small number of cycle tests. Slight mechanical adjustments were necessary in order to provide the full range of travel. The calibration and cycle tests were then repeated.

The system was then analyzed using an alignment telescope. The image of a cross-hair was projected onto the gimbal mirror, and the reflected image was examined as the gimbal was tipped. The reticle was able to be read to an accuracy of about 0.1 reticle unit, which corresponded to a mirror move of approximately 150 steps. (The exact scale was different for both axes, because of geometric factors.)

The mirror was offset by 100000 steps and then returned to its home position. This was repeated 20 times. There appeared to be a slight drift in the position, that was approximately 0.1 or 0.2 reticle units.

The mechanism was tested at cryogenic temperatures. One axis failed to move at all at cryogenic temperatures. The other axis failed also after further cold testing. The problem was eventually isolated, and repaired.

The previous calibration and test procedure was repeated, but the slight drift in position was not observed.

The mechanism was then cooled. The mirror was offset by 100000 steps and then returned to the home position. The cross-hair returned exactly to the zero position to the accuracy of observation.

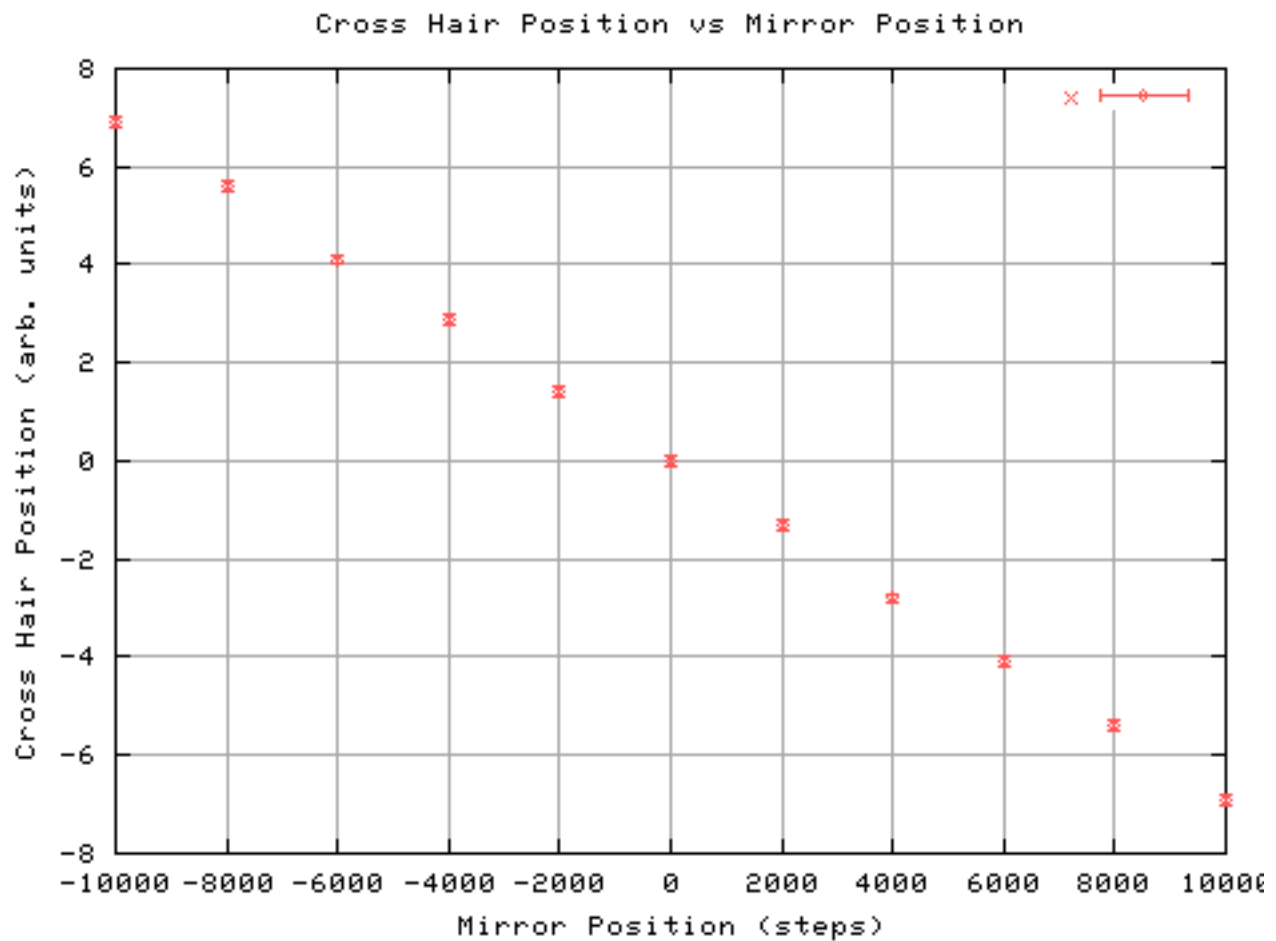
The mechanism was calibrated, and the datum / diagnose function was repeated three times for each axis. Each axis was cycle tested for five cycles.

The mechanism was then moved to a series of points, and the reticle position of the cross hair was then read. The following data was recorded. Note that no corrections were made for any slight distortions which might have been caused by the optics of the test system. The data was found to be linear, with any discrepancies easily explained by the limitations of the test setup.

TABLE 1.

x Axis Cross Hair Position as a Function of Mirror Position

Mirror position (ministeps)	Cross hair position (Reticle units)	Cross hair uncertainty (Reticle units)
-10000	6.9	0.1
-8000	5.6	0.1
-6000	4.1	0.1
-4000	2.9	0.1
-2000	1.4	0.1
0	0.0	0.1
2000	-1.3	0.1
4000	-2.8	0.1
6000	-4.1	0.1
8000	-5.4	0.1
10000	-6.9	0.1

FIGURE 1. x Axis Cross Hair Position vs Function of Mirror Position**TABLE 2.** y Axis Cross Hair Position as a Function of Mirror Position

Mirror position (ministeps)	Cross hair position (Reticle units)	Cross hair uncertainty (Reticle units)
-10000	7.8	0.1
-8000	6.0	0.1
-6000	4.7	0.1
-4000	3.0	0.1
-2000	1.5	0.1
0	0.0	0.1
2000	-1.5	0.1
4000	-3.1	0.1
6000	-4.5	0.1

Mirror position (ministeps)	Cross hair position (Reticle units)	Cross hair uncertainty (Reticle units)
8000	-6.0	0.1
10000	-7.5	0.1

FIGURE 2. y Axis Cross Hair Position as a Function of Mirror Position

